

QTM 385 – Experimental Methods

Assignment 08

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Instructions

This week's assignment focuses on interference and spillovers in field experiments. Interference occurs when the treatment status of one unit affects outcomes for other units, violating the Stable Unit Treatment Value Assumption (SUTVA). Please submit your assignment in PDF or HTML format.

If you have any questions about the assignment, feel free to email me at danilo.freire@emory.edu.

Good luck!

Questions

1. National surveys indicate that college roommates tend to have correlated weights. The more one roommate weighs at the end of the freshman year, the more the other freshman roommate weighs. On the other hand, researchers studying housing arrangements in which roommates are randomly paired together find no correlation between two roommates' weights at the end of their freshman year. Explain how these two facts can be reconciled.

National surveys showing that college roommates tend to have correlated weights likely reflect a selection bias. In many cases, students are allowed to choose their roommates or are matched based on lifestyle preferences. As a result, individuals with similar health habits, eating patterns, or physical activity levels may end up rooming together. This natural pairing process can lead to roommates starting off with similar body types or developing similar routines throughout the year, reinforcing any initial similarities and resulting in correlated weights by the end of freshman year.

In contrast, studies that involve randomly assigned roommates eliminate this selection effect. When pairings are made at random, roommates do not necessarily share similar lifestyles or

health behaviors from the outset. Without this built-in similarity, there's less chance for their habits to converge meaningfully over time, and as such, researchers observe no significant correlation in weight between the two by the end of the year. These findings suggest that the correlation seen in national surveys is not due to roommates influencing each other's weight but rather due to the types of roommates students tend to choose.

In class, we discussed an example where a field experiment was conducted to assess the impact of establishing a clinic in villages. The experiment is described on pages 260-264. The tables with the villages' potential outcomes and the probabilities of receiving treatment and spillovers are described on page 262 and shown below:

Spatial location of five villages along a line

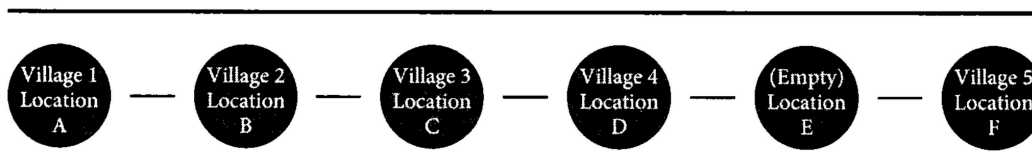


Table 1: Potential outcomes

Village	Untreated (Y_{00})	Adjacent village treated (Y_{10})	Treated (Y_{01})
A	0	2	0
B	6	2	10
C	0	4	4
D	6	6	6
F	6	NA	3

Table 2: Probability of receiving treatment, spillover, and control

Village	A	B	C	D	F	Pr(control)	Pr(spillover)	Pr(treatment)
1	Y_{01}	Y_{10}	Y_{00}	Y_{00}	Y_{00}	0.6	0.2	0.2
2	Y_{10}	Y_{01}	Y_{10}	Y_{00}	Y_{00}	0.4	0.4	0.2
3	Y_{00}	Y_{10}	Y_{01}	Y_{10}	Y_{00}	0.4	0.4	0.2
4	Y_{00}	Y_{00}	Y_{00}	Y_{10}	Y_{01}	0.6	0.2	0.2
5	Y_{00}	Y_{00}	Y_{00}	Y_{00}	Y_{01}	0.8	0	0.2

Using these data, calculate the following quantities:

1. Estimate $E[Y_{01} - Y_{00}]$ for the random assignment that places the treatment at location A.